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# The Role of Cognitive Development and Socialization in the Initial Development of Team Loyalty

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*Using Piaget's (1970) theory of cognitive development, the present study examines when children first begin to demonstrate team loyalty. An interview and testing protocol was administered to children aged 5–6 and 8–9. Preoperational, transitional, and concrete operational children were capable of demonstrating a psychological commitment to a favorite team that was resistant to change, but not the behavioral consistency indicative of loyalty. The current study demonstrates that children form preferences for sports teams early in life and that they are capable of forming a commitment to a sports team as young as age 5. The present study is one of the first efforts to include a transitional phase in the assessment of cognitive development. Results from the interviews showed that fathers were the most influential socializing agent relative to introducing children to sports teams and that the gender stereotype associating sports with males was prevalent among children in both age groups.*

**Keywords** team loyalty, cognitive development, socialization, psychological commitment

Sports spectating represents a predominant form of leisure behavior in contemporary society. Previous research with a representative national sample has demonstrated that 73% of adults in the United States are either very interested or fairly interested in watching sports (Lieberman, 1991). In 1996, attendance at professional baseball, basketball, football, and hockey games in the United States exceeded 110 million (Badenhausen, Nikolov, Alkin, & Ozanian, 1997). Along with using leisure time to attend games, people enjoy watching broadcasts of sporting events. Between 1960 and 1988, the total number of hours allotted to sports programming by NBC, CBS, and ABC increased by 500% (Madrigal, 1995). The increase in sports programming is even more dramatic, considering that the preceding figures do not include the Entertainment and Sports Programming Network, Fox Sports, and other 24-hr sports programs available today.

The interest of many spectators goes beyond investing free or non-working time attending and watching sporting events. Sports spectators also form psychological connections to specific teams that are persistent and resistant to change (Kolbe & James, 2000; Trail & James, 2001). Kolbe and James found that people buy season tickets and attend games over time (in some cases for more than 30 years) and that they think of themselves as part of the team (demonstrating a strong psychological commitment), not just as individuals who watch games. This behavioral consistency (attending multiple games) and psychological commitment suggest that many sports spectators are loyal to specific teams.

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Considering the predominance of sports spectating as a form of leisure behavior and the loyalty that people have to sports teams, the question arises as to how people initially become interested in and develop loyalty to specific teams. The current state of research on loyalty helps us to characterize the behavioral aspect of loyalty and to assess whether people have a psychological commitment to sports teams (Backman & Crompton, 1991; Day, 1969; Howard, Edington, & Selin, 1988; Jacoby & Chestnut, 1978; Pritchard, Havitz, & Howard, 1999; Pritchard, Howard, & Havitz, 1992). Yet, research has not addressed when people develop or what influences the development of such strong psychological connections to sports teams.

Previous research on loyalty has focused largely on trying to understand what loyalty is and on how to measure it (see Jacoby & Chestnut, 1978, for a summary of early research and Pritchard, 1992, for a summary of later work). More recently, Iwasaki and Havitz (1998) proposed that research should shift from looking at what loyalty is or just measuring the dimensions of loyalty to focusing on better understanding the mechanisms and conditions of how and why loyalty develops.

Iwasaki and Havitz (1998) offered a conceptual model to help explain the relationship among involvement, psychological commitment, and loyalty. The model was proposed as a resource for “assisting researchers and practitioners in thinking about the mechanisms and conditions of why and how consumers develop their loyalty to brand(s)” (p. 273). The authors suggested that research on loyalty should move from a focus on conceptualizing the topic to better understanding how loyalty develops and what influences such development. The current study is one of the first efforts to use the proposed model to investigate when an individual is first capable of demonstrating loyalty to a sports team in an effort to move the study of the construct from measuring effects (i.e., repeat behavior) to better understanding what influences an individual to make a psychological commitment that is resistant to change.

## Background

### *Preference Formation*

The conceptual framework proposes that the development of loyalty begins with involvement, a “state of motivation, arousal or interest,” in a particular leisure activity (Iwasaki & Havitz, 1998, p. 260). Before individuals develop a psychological commitment to a team, they must first decide that they like one team better than others. The initial attraction to or interest in a sports team may be thought of as having a preference for a specific team.

Preference for specific leisure activities is a topic that has received considerable attention from researchers. Previous research has focused on identifying leisure preferences among adults (Driver, 1976; Knopf, 1983); determining the types of recreation experiences people prefer (e.g., escape, stimulation seeking; Ballman, Knopp, & Merriam, 1981; Manfredo, Driver, & Tarrant, 1996; Manfredo & Larson, 1993); identifying the relationship between experience preferences and personality traits (Driver & Knopf, 1977) and values (Manfredo, Sneegas, Driver, & Bright, 1989); assessing the stability of preferences (Crawford, Godbey, & Crouter, 1986); and comparing leisure preferences as they relate to race, social class, gender, and socioeconomic levels (Floyd, Shinew, McGuire, & Noe, 1994; Shinew, Floyd, McGuire, & Noe, 1995, 1996).

Very little effort has addressed when and how preferences form, and no research has investigated the progression from preference for to commitment to a leisure activity. Insight into the formation of leisure preferences was offered by Crawford et al. (1986) in their discussion of preference stability. Citing the work of Kleiber and Durkin (1985), Crawford et al. noted that “leisure preferences may be the by-product of personal histories resulting

from learning and socialization” (p. 96). Iso-Ahola and Mannell (1985) also emphasized that leisure choices are socially affected and that people revise their leisure goals, patterns, and habits relative to the social environment. Crawford et al. pointed out that the formation and stability of leisure preferences are best understood through examining an individual’s psychological orientation and the influence of environmental factors (socializing agents). In other words, the formation of leisure preferences is thought to occur through the process of leisure socialization.

### ***Leisure Socialization***

Hoff and Ellis (1992) presented a review of the perspectives explaining leisure socialization in their study of the influence of leisure socialization agents on leisure self-efficacy. One perspective describes leisure socialization as a process through which individuals acquire knowledge, attitudes, values, skills, and motives about leisure (Iso-Ahola, 1980). This perspective suggests that socialization into leisure is a continuous pattern of change beginning with the earliest interactions of childhood, continuing through the pivotal periods of adolescence and young adulthood and on through maturity.

Within this socialization process Bandura (1986, p. 415) proposed that family contacts provide the initial and perhaps most lasting examples of modeled behavior. Eventually, family contacts are augmented by contacts with peers, the school community, and ever widening contacts with other sources of influence. (Hoff & Ellis, 1992, p. 115).

A second perspective examines the process through which people are socialized into particular recreation activities (Burch, 1969; Kelly, 1972; Kleiber & Kelly, 1980). One of the important findings from these studies is that family structures may provide the primary path for socialization into particular patterns of recreation participation. Other studies (Crandall, 1979; Fedler, 1984) evaluated the role social interaction plays in influencing leisure choices. The importance of social interaction has been noted in these studies, along with the influence of norm reference groups, namely family and friends (Fedler, 1984). Examining the role of socializing agents on the formation of leisure preferences is one method of investigating the influence of social-situational moderators on the development of team loyalty.

### ***Sport Socialization***

Research in a sports context has examined the process of socialization into sports participation (Kenyon & McPherson, 1973; McPherson, 1976; Snyder & Spreitzer, 1989). Kenyon and McPherson reported that before high school, individuals are encouraged to participate in sports primarily by family members and to a lesser extent by friends. In high school, individuals receive greater reinforcement from coaches and peers than from family members. For college athletes, the greatest amount of reinforcement comes from peers. These findings suggest that a variety of agents can have an influence on a child’s interest and involvement in sports and also that various socializing agents exert a differential influence over time.

McPherson (1976) analyzed the contribution of four sources of sports socialization (family, peers, school, and community) to a propensity for sports consumerism. Building on the theoretical basis proposed by Bandura (1969), McPherson proposed that socialization into sports was similar to socialization in general. From a study of high school students, McPherson found that peer groups were most influential for males, and parents were most influential for females.

Previous research on leisure and sports socialization has indicated that the process occurs throughout the life cycle, with family and peers exerting an important influence

on an individual, particularly early in life. The implication for the development of team loyalty is that preferences for sports teams may form early in life as a result of socialization from family and to a lesser degree from friends. It is interesting to note that the majority of research on leisure and sports socialization has included adults, university students, and, in a few cases, high school students. Research has also focused on participation in activities and has not examined socialization into sports spectating (with the exception of propositions offered by McPherson, 1976). To better understand when and how team loyalty develops, it is important to examine the formation of preferences and commitment beginning in childhood.

Research in consumer behavior has emphasized the importance of studying the development of loyalty early in life. Findings have shown that preferences can form very early in life (Bahn, 1986) and that commitments made early in life tend to endure (Guest, 1964). Guest reported that people who identified a strong attachment to a specific product maintained their attachment over a 20-year period. Using Piaget's (1970) cognitive developmental stage theory, Bahn examined whether children had developed the cognitive abilities to discriminate between and form preferences for specific brands. Preoperational children (ages 2–7) categorized brands using a single dimension (e.g., physical aspects), whereas concrete operational children (ages 7–12) categorized brands using multiple dimensions (e.g., physical aspects, uses, and abstract attributes). The results indicated that children at both stages of cognitive development had distinct preferences for cereals and beverages, but preferences among concrete operational children were based on multiple dimensions (vs. one) and were more stable (resistant to change).

Although Bahn (1986) did not assess how preferences were formed, his work introduced a second important dimension to consider: level of cognitive development. The preceding discussion of preference formation and socialization highlights the importance of examining the influence of socializing agents, particularly family and friends, on the development of commitment and loyalty. At the same time, the conceptual framework emphasizes the importance of understanding the influence of personal moderators. The results reported by Bahn suggest that in order for loyalty to develop, to make a psychological commitment to a specific team, there must be a minimum level of cognitive development.

### *Cognitive Development*

Cognitive development is the process by which people learn to recognize the larger world around them, to process information, and to make distinctions between different objects and situations (Singer & Revenson, 1978). It is the means by which individuals learn to value or form preferences for different objects (e.g., having a favorite team) and potentially make a psychological commitment to an object (e.g., the Chicago Cubs). Preferences form early in life (Bahn, 1986; Guest, 1964), but there is a question as to how and when children progress from having a preference for to developing loyalty to an object.

Considering that loyalty is reflected in psychological commitment—which includes volition, cognitive complexity, and resistance to change—and behavioral consistency (Pritchard et al., 1992), it is reasonable to expect that a person does not become loyal to a team on a whim. Loyalty is the result of making cognitive distinctions between objects and forming psychological connections that are persistent and resistant to counterpersuasion. A certain amount of thought is involved in terms of comparing teams, evaluating differences between and among teams, and forming an attachment to a particular team. Even if an individual's choices are constrained by availability (they only get to see one team's games during the season on television) or perceived acceptability (their family and friends think there is only one team to support), becoming loyal to a team requires a minimum level of thought and reasoning, a minimum level of cognitive development.

### ***Piaget's Theory***

A well-known approach to assessing cognitive development is based on the work of Jean Piaget (1970). "Piaget's theory is the most comprehensive available theory of cognitive development" (Lutz & Sternberg, 1999, p. 276), and since the mid-1960s, the nature of children's thinking and learning has been influenced by Piaget's idea (Siegler, 1991; Sutherland, 1992; Wood, 1988). Piaget's theory presents a detailed and age-related account of four broad phases involved in the development of intellect. His work provides a possible explanation as to when and how a child learns or develops specific forms of knowledge and understanding. The four phases proposed by Piaget are

Sensorimotor → Preoperational → Concrete operations → Formal operations  
Development                      Thought

Of interest to the initial development of team loyalty are the phases of preoperational thought and concrete operations. Within these two phases, Piaget (1970) proposed that children begin to (a) distinguish between groups and categories (e.g., distinguish between different sports and different teams), (b) demonstrate stable preferences, and (c) include affective components in their decision-making process (emotions, values, and feelings contribute to decisions). Such characteristics are important for comparing and evaluating differences between and among sports teams and are necessary precursors for loyalty to a specific sports team to develop.

### ***Research Questions***

The lack of research on when and how loyalty develops suggests that an important first step is to establish a fundamental understanding of how children become aware of sports teams and begin to form a psychological commitment to a specific team that is resistant to change. The characteristics proposed by Piaget (1970) would suggest that children in the preoperational phase of development would be capable of forming preferences for a favorite team but would lack the cognitive complexity to develop a psychological commitment to a sports team. Children characterized by concrete operational thought, however, should have the cognitive complexity to form a psychological commitment that is resistant to change.

The review of leisure and sports socialization research suggests that for children characterized by preoperational thought (approximately 4–7 years of age) and concrete operational thought (approximately 8–12 years of age), family members and friends would likely exert the most influence on children forming attachments to sports teams. It is not clear, however, which family members would exert more or less influence and to what extent friends may influence children relative to family members. Using the conceptual framework proposed by Iwasaki and Havitz (1998), the following research questions were proposed as a first step to establish a better understanding of when and how loyalty to a sports team may develop relative to a child's level of cognitive development and the socializing influence of family and friends.

*Research Question 1:* Are children at preoperational and concrete operational levels of cognitive development capable of demonstrating the psychological commitment and behavioral consistency that characterize loyalty to a sports team?

*Research Question 2:* Do family members exert a greater relative influence than friends on a child's attachment to a favorite sports team at the preoperational phase of development?

*Research Question 3:* Do family members exert a greater relative influence than friends on a child's attachment to a favorite sports team at the concrete operational phase of development?

**Method**

A qualitative study was conducted with two groups of children ( $N = 50$ ) aged 5–6 ( $n = 25$ ) and 8–9 ( $n = 25$ ). Age groupings provide only a cursory indication of level of cognitive development, but previous research (Siegal, Waters, & Dinwiddy, 1988) has demonstrated that children under age 7 are likely to be characterized by preoperational thinking and children between ages 7 and 12 are likely to be characterized by concrete operational thinking. A convenience sample was used because of manpower and budgetary constraints. An elementary school in a large city in the Midwest was contacted about participation in the project. After obtaining permission from school administrators, parents of children in kindergarten and third grade were contacted. Consent was obtained from parents for children to participate in the project.

I spent time assisting with class activities for both groups so that the children could get to know and be comfortable talking with me. I conducted the interviews in a classroom at the school, a familiar environment for the children. From the child's perspective, the interview involved talking about some of his or her favorite objects and why she or he liked particular objects.

A clinical observation technique was implemented using a protocol (described below) to interview children thought to be at different levels of cognitive development. The protocol included the administration of three conservation tasks drawn from previous research (Siegal et al., 1988; Wood, 1988) to assess level of cognitive development and a series of questions assessing if a child was loyal to a sports team. Questions were also included to examine the extent to which different socializing agents influenced a child's attachment to a favorite team.

Interviews were tape recorded and transcribed, and the transcriptions were entered as data files into *Q.S.R. Nudist* (1996). The *Nudist* software is designed for qualitative analysis, allowing the researcher to identify patterns of responses and to make comparisons between groups. Key words, phrases, and questions selected from the interview protocol (e.g., *favorite*, *favorite team*, *Why do you like your favorite team?*) were used to assess the children's responses. Using level of cognitive development as a grouping variable, it was possible to create separate files for each behavioral and attitudinal element that included a designated phrase or question. For example, the phrase *favorite team* was set up as a search string. The *Nudist* software was used to select out each conversation block from all the transcripts in which the phrase *favorite team* occurred. The software made it possible to isolate all responses pertaining to having a favorite team. Bringing together all the responses to a single question and all the responses that included a specific word or phrase, I was able to measure the frequency of responses and make comparisons between preoperational and concrete operational children.

**Interview Protocol**

The interview protocol was modeled on the clinical observation techniques developed by Piaget (Piaget & Inhelder, 1969) and used by child development specialists (see Siegler, 1991, for a synopsis). The interview protocol is not a formalized test or series of questions, it is a process of assessing a child's level of reasoning about a particular concept. The interviewer has established question areas or topics. The interview is a dynamic exchange between the interviewer and the child that provides information about a child's thinking,



reasoning, and understanding. The interview protocol protected against leading or cueing a child's responses and also provided the opportunity to clarify responses.

### ***Level of Cognitive Development***

Piaget's (1970) theory of cognitive development has sparked much discussion and debate. Of interest to the current study is the work on the preoperational and concrete operational phases, particularly concerns with the conservation tasks used to distinguish between the two phases. Two of the primary concerns have been the manner in which questions are phrased and the wording of questions. One manner in which questions are phrased attempts to "catch" a child. For example, when administering the tasks the set of answers children are given to select from may or may not include the correct answer. Research has found that using questions that include the correct response enables children to recognize that changing appearances does not necessarily change number, volume, or mass (Bryant & Trabasso, 1971; Gelman, 1972) at earlier ages than Piaget suggested.

The concern with the wording of questions has been the use of language that children may not understand. Through the use of children's toys, language that children understand, and working in familiar environments, McGarrigle and Donaldson (1974) and McGarrigle, Grieve, and Hughes (1978) found that children were able to demonstrate concrete operational thinking earlier than Piaget (1970) proposed.

Investigations of the initial ideas suggest that although Piaget (1970) likely underestimated younger children's reasoning abilities, a modified version of his ideas is plausible. Sutherland (1992) argued that the "phenomenon of operational thinking does exist and that it is sometimes attained at an earlier age than Piaget indicated" (p. 114). That children may demonstrate behaviors and cognitions earlier than proposed by Piaget does not detract from the differences between stages. The implication for the current study is that children at a younger age (e.g., 6 years) may demonstrate characteristics of concrete operational thinking and, potentially, team loyalty.

Age ranges are used as relative indicators; of primary importance is distinguishing between the phases of development. Research has demonstrated that children are able to conserve using a single-question task in which a transformation is performed in view of a child, followed by a question asking if two objects are the same (Perner, Leekam, & Wimmer, 1986; Siegal et al., 1988). Siegler (1991) also proposed that Piaget's (1970) fundamental concepts—including concepts of space, time, and number—do provide insight into children's thinking that merits continued examination. To appropriately use the conservation tasks, the answer set should include the correct response and the questions should be asked using language that is clear to children.

Using the proposed modifications, the first phase of the interview assessed a child's level of cognitive development based on two criteria: (a) correct responses to two of the three cognitive development tasks and (b) demonstration by the child through his or her explanation that she or he understood the conservation concept. Three cognitive tasks were used to assess conservation of number, mass, and volume. In each task, a child observed two identical conditions; a transformation was performed in front of the child, and a question was then asked about the objects. After responding to each question, a child was asked to explain why she or he gave a particular answer. Simply responding correctly to a question does not necessarily indicate that a child understands a concept. By telling why he or she gave a particular response, a child demonstrates understanding of a conservation concept.

For conservation of number, a child was shown two identical rows of six pennies; while observing the rows of pennies, one row was spread out so that it appeared longer. The child was then asked if she or he thought there were more pennies in the first row, more pennies in the second row, or the same number of pennies in both rows. For conservation of mass,

a child was shown two identical balls of Play-Doh; while observing the two balls, one was flattened out. The child was then asked if she or he thought there was more Play-Doh in the ball, more in the flat object, or about the same amount in each. For conservation of volume, a child was shown two identical containers of colored water. In front of the child, the water from one container was poured into a taller, thinner container. The child was then asked if she or he thought there was more water in the shorter container, more water in the taller container, or about the same amount of water in each container. Correct responses for each task were that the two conditions were identical. Correct reasoning in the respective tasks was that spacing did not change the number in each row, changing the shape did not change the amount in each object, and a tall, narrow container could hold the same amount of water as a short, wide container. Responses to the conservation tasks were used to classify the children on the basis of their level of cognitive development.

### ***Demonstration of Loyalty***

The second phase of the interview assessed whether a child had a favorite sports team and, if so, whether she or he was loyal to that team. Questions were developed to assess psychological commitment (cognitive complexity, resistance to change, volition) and behavioral consistency. No previous research has examined children's loyalty to sports teams, so there was no standard on which to model the interview protocol. Questions assessing behavioral consistency were developed from the characteristics commonly associated with loyal fans (McPherson, 1976; Smith, Patterson, Williams, & Hogg, 1981). Children may not demonstrate fully developed characteristics of team loyalty, but it is reasonable to expect that if loyalty were developing distinct behaviors would be evident. Drawing from the work of McPherson and Smith et al., five behavioral elements were measured: (a) attending a favorite team's games, (b) watching a favorite team on television, (c) reading about a favorite team, (d) talking about a favorite team, and (e) owning team-related merchandise.

Assessment of a child's psychological commitment was based on questions developed from the work of Pritchard (1992). Cognitive complexity was assessed by asking a child to discuss why she or he liked a particular team. Individuals with higher levels of commitment should have more complex cognitive structures (i.e., ideas that have a rich content or multiple ideas associated with an object) and stronger attachments to an attitude object (e.g., a sports team; Freedman, 1964). Because research has not yet measured a child's commitment to a sports team, the current study used two criteria to gauge cognitive complexity: (a) a child's ability to give at least one "complex" reason (complex being more than "because I like them") or (b) a child's ability to give multiple (two or more) reasons for having a favorite team. These criteria were drawn in part from Bahn's (1986) finding that preoperational children distinguished between favorite items using one attribute, whereas concrete operational children used multiple attributes.

Resistance to change challenged the strength of attachment by asking a child if she or he would continue to like a specific team even if the team lost all of its games or if there were some change in the team (e.g., a team moved to another city or star players left the team). Children who indicated that they would continue to like a favorite team when "challenged" were characterized as resistant to change.

Volition was measured by asking children if their family and friends had a favorite sports team and whether the children would continue to like their favorite team even if others (family and friends) did not like that particular team. Children knowing that others liked a specific team yet choosing a different team as their personal favorite demonstrated volition. Children not liking a team if their friends or family did not like that team showed a lack of volition. Volition was a difficult concept to measure. It involves free choice and self-responsibility for one's preferences (Pritchard, 1992). The challenge was measuring

whether a child's preference for a specific team was a personal choice or based on other extrinsic influences. Evidence of volition was also sought from a child's reasons for liking a favorite team (e.g., because his or her father liked the team vs. personal reasons for liking a team). Table 1 outlines the elements measured for each dimension of loyalty and illustrates the question content used for each dimension.

The interviews were conducted in a city that had no top-level (National Football League [NFL], Major League Baseball [MLB], National Basketball Association [NBA], or National Hockey League) sports team, but was home to a new Major League Soccer (MLS) team, several minor league teams, and university teams competing in Division I-A sports. To assess team loyalty in the context of sports spectating, I determined that local (minor league)

**TABLE 1** Measures of Team Loyalty

| Loyalty dimension & elements examined | Question content                                                                                                                                                                       | Questions derived from                                                         |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Behavioral                            |                                                                                                                                                                                        | Characteristics of highly committed fans (McPherson, 1976; Smith et al., 1981) |
| Attendance                            | Do you attend your favorite team's games?                                                                                                                                              |                                                                                |
| TV viewing                            | Do you watch your favorite team on television?                                                                                                                                         |                                                                                |
| Reading                               | Do you read about your favorite team?                                                                                                                                                  |                                                                                |
| Talking                               | Do you talk with others about your favorite team?                                                                                                                                      |                                                                                |
| Team merchandise                      | Do you own team-related merchandise?                                                                                                                                                   |                                                                                |
| Attitudinal                           |                                                                                                                                                                                        | Psychological commitment (Pritchard, 1992)                                     |
| Cognitive complexity                  | Why is "Team A" your favorite team (quantity and specificity of reasons)?                                                                                                              |                                                                                |
| Volition                              | Do you like "Team A" because your dad likes "Team A?"<br>Do you like "Team A" because your friends like "Team A?"<br>Do you know your [family member, friend's] favorite [sport] team? |                                                                                |
| Resistance to change                  | Would you still like a team even if they lost all their games?<br>Would you still like a team when there are changes (team moves, players leave, etc.)?                                |                                                                                |

*Note.* Question content reflects the types of questions asked, not the exact phrasing used during the interviews.

professional teams, other professional teams and college teams would be considered for the study of team loyalty. The children were not required to talk about any particular sport or level of sport. Children were free to talk about sports teams in the context of high school sports, or even youth league sports. For the purpose of the study, however, team loyalty was assessed in the context of professional and collegiate sports.

### *Socializing Influences*

A third part of the interview examined the influence of three socializing agents—parents, siblings, and friends. Children were asked whether their family and friends had a favorite sports team. If a child knew whether family members and friends have a favorite team, that would indicate that family members and friends had talked about a team or in some way communicated their attachment to a specific team (e.g., watching a team on television, attending games, etc.). If a child's favorite team was the same as a particular socializing agent, it is reasonable to consider that the agent had some influence on the child's developing an interest in the same team. If a child had a favorite team that was different from family members or friends, or if a child did not know whether family members or friends had a favorite team, it would seem reasonable to suggest that the socializing agent(s) had little influence on the child selecting a favorite team.

A child's knowing that he or she has the same favorite team as a father or brother does not indicate to what extent either influenced the child's attachment to a specific team. The influence of the socializing agents was further examined by asking each child if she or he attended or watched games with others (parents, siblings, or friends), and whether they talked with others about their favorite team. Time spent attending or watching games, or talking about a favorite team, would provide opportunities for a socializing agent to teach a child the perceived importance of supporting a particular team, and potentially help foster a child's attachment to a specific team.

### *Parental Survey*

The final element of the study was a parental survey. Parents were asked to provide information about their interest in sports in general and whether they had a favorite college or professional sports team. The questionnaire also included questions about whether parents watched their favorite team on television with their children and if they talked about a favorite team with their children. These items were included to clarify the extent to which parents did interact and potentially influence the development of loyalty to a specific team.

### *Pilot Study*

A pilot study was conducted to test the interview protocol. A convenience sample of children ( $N = 7$ , 3 girls and 4 boys) ranging in age from 4 to 11 was used for the pilot test. A range of ages was used in order to test the protocol with children thought to be at different levels of cognitive development. It was important to demonstrate that the conservation tasks would distinguish between preoperational and concrete operational children and to ensure that children at different ages would understand the questions. Parental consent was obtained to interview the children in their homes; I conducted the interviews over the period of 1 month.

Results of the pilot study were consistent with previous use of the conservation tasks. The four older children (aged 8–11) were characterized as concrete operational and the three younger children (aged 4–7) as preoperational. Children in both groups were able to identify a favorite team, and there were differences relative to the behavioral and attitudinal dimensions of team loyalty. Some but not all of the behavioral elements were demonstrated. Most noticeable among the preoperational group was the lack of reading about a favorite

team due to the fact that the children were just learning how to read. Only the 4-year-old had difficulty answering some of the questions in the protocol. The questions that were difficult for the 4-year-old were modified so that they would be easier to understand. A review of the transcripts provided direction for the sequencing of questions—begin with questions about what a child does for fun, followed by questions about favorite objects (e.g., food, games, etc.), and conclude with questions about sports teams.

An additional element relative to level of cognitive development emerged through the pilot study. One criticism of Piaget's (1970) theory not cited earlier is disagreement with the idea that cognitive development occurs in a fixed sequence of discontinuous stages. Previous research (Bahn, 1986; Perner et al., 1986; Piaget & Inhelder, 1969; Siegal et al., 1988) has classified children as preoperational or concrete operational on the basis of the results of the conservation tasks. Using both the correct response and the correct reasoning criteria to characterize a child's cognitive development means that if a child recognizes the transformation but has difficulty explaining why the objects are the same, she or he would be classified as preoperational. It can be argued that a child truly characterized by preoperational thought would not be able to conserve number, mass, or volume, meaning she or he should not recognize or be able to explain why two objects that appear dissimilar are the same. A child who recognizes that two objects that look dissimilar are actually the same, and who can explain the conservation tasks, would be characterized as concrete operational.

The question arises, though, about how to characterize children who recognize the similarity between objects but have difficulty explaining why they are the same, or who offer explanations that do not reflect conservation of number, mass, or volume. Previous research would conclude that such children are not concrete operational; but should they be characterized as preoperational? Recognizing the similarity between objects seems to indicate that a child is developing, but she or he may not have fully achieved concrete operational thought. In other words, a child may demonstrate that she or he is transitioning between preoperational and concrete operational thinking.

The difficulty with characterizing a child's level of cognitive development does not diminish the value of assessing preoperational and concrete operational thought. The response to a conservation task and the reasoning a child uses to explain the task provide a good foundation for assessing level of cognitive development. Rather than discounting the ideas of preoperational and concrete operational thinking altogether, I chose to characterize children as preoperational, transitional, or concrete operational (criteria described below) in order to better understand when a child is capable of demonstrating team loyalty. Supported by the results of the pilot study, interviews with 50 children in two age groups (5–6 and 8–9) were conducted.

## Results

### *Sample Characteristics*

The sample characteristics of the children interviewed are presented in Table 2. The group was evenly split between boys and girls; two thirds were Caucasian; 86% lived in a home with both parents. A cross-tabulation of age and personal characteristics found one significant difference based on ethnicity,  $\chi^2 (3, N = 50) = 11.30, p < .01$ . Within the 5–6-year-old group, there were no 6-year-old African Americans.

### *Cognitive Development*

Children were asked, for each conservation task, if the objects being compared were the same or different; a child scored 1 point if she or he gave a correct response that the two objects were the same. A second point was scored if a child was able to explain why the

TABLE 2 Sample Characteristics

| Characteristic         | Age (in years) |    |    |   | Total | %  | $\chi^2$ | <i>p</i>  |
|------------------------|----------------|----|----|---|-------|----|----------|-----------|
|                        | 5              | 6  | 8  | 9 |       |    |          |           |
| Gender                 |                |    |    |   |       |    | 1.348    | <i>ns</i> |
| Male                   | 7              | 7  | 7  | 4 | 25    | 50 |          |           |
| Female                 | 7              | 4  | 10 | 4 | 25    | 50 |          |           |
| Ethnicity              |                |    |    |   |       |    | 11.300   | <.01      |
| Black/African American | 8              | 0  | 7  | 1 | 16    | 32 |          |           |
| White/Caucasian        | 6              | 11 | 10 | 7 | 34    | 68 |          |           |
| Household              |                |    |    |   |       |    | 4.861    | <i>ns</i> |
| Single parent          | 4              | 2  | 1  | 0 | 7     | 14 |          |           |
| Dual parent            | 10             | 9  | 16 | 8 | 43    | 86 |          |           |

*df* = 3; *N* = 50.

transformation witnessed did not alter the number, mass, or volume of the objects. Children who thought the objects were different based their reasoning on the objects’ appearance. For example, a child stating that “the longer row has more pennies” was not able to conserve number. If a child said that one ball had more Play-Doh because it was bigger or wider, she or he was not able to conserve mass. A child who said that one container had more water because the water level in one of the containers was lower or higher was not able to conserve volume.

Children who correctly responded that the objects being compared were the same and who were able to explain why the transformation did not alter the objects for at least two of the tasks were characterized as concrete operational. By meeting these criteria, a child would score a minimum of 4 points. Children who correctly responded on two of the conservation tasks that the objects compared were the same, but whose reasoning was not based on the transformation, were characterized as transitioning between preoperational and concrete operational thinking. Children meeting these criteria would score 2 or 3 points.

The children characterized as transitional gave only two reasons to explain the similarity between objects. The most frequent response was that two objects looked the same. After the transformation, the appearance of the objects was clearly different. The response that two objects looked the same was interpreted to mean that a child was remembering the condition before the transformation, which is indicative of some abstract thinking associated with concrete operational thought. The children did not seem to have progressed, however, to the point where they were processing in their mind the original condition and the transformation to conclude that there was no change in the objects.

The second reason given to explain why two objects were the same occurred only with the conservation of number task. Instead of saying that two rows of pennies looked the same, or that the spacing did not alter the number, children counted the pennies and concluded that each row had the same number. The counting explanation was interpreted to mean that a child did recognize the transformation, but instead of comparing the spacing of the pennies abstractly (i.e., comparing in his or her mind the spacing of the rows before and after the transformation), a child counted the pennies in each row. Counting the pennies suggested that a child focused on his or her immediate perceptions, which is indicative of preoperational thinking. At the same time, counting the pennies showed that a child did acknowledge the transformation, signaling some transition from preoperational to concrete operational thinking.

**TABLE 3** Frequency of Conservation Task Scores and Stage of Cognitive Development

|                                      | Age (in years) |   |   |   |   |   |   |   | Total | %  | $\chi^2$            | <i>p</i> |  |
|--------------------------------------|----------------|---|---|---|---|---|---|---|-------|----|---------------------|----------|--|
|                                      | 5              |   | 6 |   | 8 |   | 9 |   |       |    |                     |          |  |
|                                      | M              | F | M | F | M | F | M | F |       |    |                     |          |  |
| Conservation task score <sup>a</sup> |                |   |   |   |   |   |   |   |       |    |                     |          |  |
| 0                                    | 1              | 3 | 2 |   |   |   |   |   | 6     |    |                     |          |  |
| 1                                    | 2              | 1 | 2 |   |   | 1 |   |   | 6     |    |                     |          |  |
| 2                                    | 1              | 1 | 1 |   |   |   |   |   | 3     |    |                     |          |  |
| 3                                    | 2              | 1 | 1 | 3 | 5 | 4 |   | 1 | 17    |    |                     |          |  |
| 4                                    |                |   |   |   |   |   |   |   | 0     |    |                     |          |  |
| 5                                    |                |   | 1 |   | 2 | 2 | 1 | 2 | 8     |    |                     |          |  |
| 6                                    | 1              |   | 1 | 1 |   | 3 | 3 | 1 | 10    |    |                     |          |  |
| Stage of development                 |                |   |   |   |   |   |   |   |       |    | 20.084 <sup>b</sup> | .01      |  |
| Preoperational                       | 3              | 4 | 4 |   |   | 1 |   |   | 12    | 24 |                     |          |  |
| Transitioning                        | 3              | 2 | 2 | 3 | 5 | 4 |   | 1 | 20    | 40 |                     |          |  |
| Concrete operational                 | 1              | 1 | 1 | 1 | 2 | 5 | 4 | 3 | 18    | 36 |                     |          |  |

Note. M = male; F = female.

<sup>a</sup>One point was awarded for a correct response to the question asked for each conservation task (number, mass, and volume). A second point was awarded for correctly explaining a particular conservation concept.

<sup>b</sup>A significant difference was found between the three groups.

The scores for the conservation tasks are reported in Table 3. Twenty-four percent of the children were characterized as preoperational (scored 0 or 1), 40% as transitional (2 or 3 points), and 36% as concrete operational (4–6 points). It should be noted that the children who scored 1 point did correctly respond to one conservation task, but they were not able to explain why the two objects were the same, not even the reason that they looked the same. I believe that this lack of reasoning reflected preoperational thinking rather than transitional thinking.

A cross-tabulation of age and cognitive development indicated that there was a significant difference,  $\chi^2(6, N = 50) = 20.08, p < .01$ , between the groups. The difference was attributable to 92% of the preoperational children being 5 or 6 years old and 78% of the concrete operational children being 8 or 9 years old. No significant differences were found between cognitive development and ethnicity or household make-up.

The distribution by age and cognitive development was consistent with the age ranges proposed for preoperational and concrete operational thinking. The characterization of four children who were 5 or 6 years old as concrete operational does illustrate that Piaget (1970) likely underestimated the age at which children may demonstrate concrete operational thinking. It is unclear why one 8-year-old was characterized as preoperational. The results could have been due to not communicating well with the child and his or her misunderstanding the conservation tasks, the child's trying to give the answers she or he thought the interviewer wanted, or the child transitioning from preoperational to concrete operational thinking.

### **Demonstration of Team Loyalty**

#### **Favorite Team**

Each child was asked a variety of questions about favorite objects (e.g., food, games, etc.), including whether she or he had a favorite sports team. Preference for a particular team

TABLE 4 Frequency of Responses to Team Loyalty Measures

| Measure & response          | Preoperational |   |       | Transitional |   |    | Concrete operational |    |   | $\chi^2$ | <i>p</i> |    |                    |                   |
|-----------------------------|----------------|---|-------|--------------|---|----|----------------------|----|---|----------|----------|----|--------------------|-------------------|
|                             | M              | F | Total | %            | M | F  | Total                | %  | M |          |          | F  | Total              | %                 |
| Have a favorite sports team |                |   |       |              |   |    |                      |    |   |          |          |    | 3.058 <sup>a</sup> | <i>ns</i>         |
| Yes                         | 4              | 0 | 4     | 33           | 9 | 4  | 13                   | 65 | 6 | 3        | 9        | 50 |                    |                   |
| No                          | 3              | 5 | 8     | 67           | 1 | 6  | 7                    | 35 | 2 | 7        | 9        | 50 |                    |                   |
| Psychological commitment    |                |   |       |              |   |    |                      |    |   |          |          |    |                    |                   |
| Cognitive complexity        |                |   |       |              |   |    |                      |    |   |          |          |    | 3.681 <sup>a</sup> | <i>ns</i>         |
| Yes                         | 2              | 0 | 2     | 17           | 7 | 2  | 9                    | 45 | 6 | 3        | 9        | 50 |                    |                   |
| No                          | 5              | 5 | 10    | 83           | 3 | 8  | 11                   | 55 | 2 | 7        | 9        | 50 |                    |                   |
| Resistance to change        |                |   |       |              |   |    |                      |    |   |          |          |    | 1.727 <sup>a</sup> | <i>ns</i>         |
| Yes                         | 4              | 0 | 4     | 33           | 7 | 4  | 11                   | 55 | 4 | 3        | 7        | 39 |                    |                   |
| No                          | 3              | 5 | 8     | 67           | 3 | 6  | 9                    | 45 | 4 | 7        | 11       | 61 |                    |                   |
| Volition                    |                |   |       |              |   |    |                      |    |   |          |          |    | 2.818 <sup>a</sup> | <i>ns</i>         |
| Yes                         | 1              | 0 | 1     | 8            | 6 | 1  | 7                    | 35 | 3 | 2        | 5        | 28 |                    |                   |
| No                          | 6              | 5 | 11    | 92           | 4 | 9  | 13                   | 65 | 5 | 8        | 13       | 72 |                    |                   |
| Commitment demonstrated     | 1              | 0 | 1     |              | 5 | 1  | 6                    |    | 1 | 2        | 3        | 20 |                    | 0.25 <sup>a</sup> |
| Behavioral consistency      |                |   |       |              |   |    |                      |    |   |          |          |    |                    | <i>ns</i>         |
| Attendance                  |                |   |       |              |   |    |                      |    |   |          |          |    |                    |                   |
| Yes                         | 2              | 0 | 2     | 17           | 3 | 0  | 3                    | 15 | 2 | 1        | 3        | 17 |                    |                   |
| No                          | 5              | 5 | 10    | 83           | 7 | 10 | 17                   | 85 | 6 | 9        | 15       | 83 |                    |                   |
| Television viewing          |                |   |       |              |   |    |                      |    |   |          |          |    | 2.194 <sup>a</sup> | <i>ns</i>         |
| Yes                         | 4              | 0 | 4     | 33           | 8 | 1  | 9                    | 45 | 2 | 2        | 4        | 22 |                    |                   |
| No                          | 3              | 5 | 8     | 67           | 2 | 9  | 11                   | 55 | 6 | 8        | 14       | 78 |                    |                   |





is not indicative of loyalty, but as proposed by Iwasaki and Havitz (1998), attraction should be an antecedent of commitment and, subsequently, loyalty. The frequency of responses reported in Table 4 shows that children in all three groups said that they did have a favorite sports team. Favorite teams included the local MLS team, other professional teams (one NBA team, four NFL teams, and three MLB teams), and the local and one nonlocal college football team.

There was no overall difference between level of cognitive development and having a favorite college or professional sports team. There was, however, a significant difference by gender within the preoperational group,  $\chi^2(1, N = 12) = 4.286, p < .05$ , and the transitional group,  $\chi^2(1, N = 20) = 5.495, p < .05$ . No girls in the preoperational group had a favorite sports team. Within the transitional group, only 1 boy out of 10 and 6 girls out of 10 did not have a favorite team.

Only 7 of the 26 children who had a favorite team were female (see Table 4). The majority of preoperational children were 5 or 6 years old, and no girls in that group had a favorite team. A question to consider is whether sports spectating is thought of as a boys' but not a girls' activity, particularly among younger children. Socialization relative to participating in sports may be more balanced across gender (Lewko & Greendorfer, 1988), but among the children interviewed there seemed to be a gender difference for sports spectator socialization.

No significant difference was found between having a favorite team and household make-up, but there was a difference within the preoperational group,  $\chi^2(1, N = 12) = 4.286, p < .05$ , on the basis of ethnicity. No African American children reported having a favorite sports team. It is not clear why this difference emerged. These children may not have had the opportunities or desire to watch college or professional sports, particularly as the only top-level sport in the community was major league soccer.

### ***Psychological Commitment—Cognitive Complexity***

Children in all three groups offered reasons that were interpreted as complex (see Table 4). Reasons thought to reflect cognitive complexity included "I like the team because they win all the time," "I like the Cubs because I played on a team called the Cubs," and "I live in Ohio and the team is from Ohio." The responses given most frequently by the children who said they did have a favorite team but who did not demonstrate cognitive complexity were "because they are my favorite" or "because I like them."

No significant difference was found between the stages of cognitive development and cognitive complexity. There was a gender difference, however, within the transitional group,  $\chi^2(1, N = 20) = 5.051, p < .05$ . More boys than girls gave complex reasons for liking a particular team. Consistent with Bahn's (1986) findings, the preoperational children gave only one reason for having a favorite team. Forty-five percent of the transitional children and 50% of the concrete operational children gave two or more reasons to explain why they liked a team. The results suggest that children do form preferences for sports teams early in life and in early stages of cognitive development. There was no significant difference relative to ethnicity or household make-up based on cognitive complexity.

### ***Psychological Commitment—Resistance to Change***

Forty-four percent of all the children interviewed demonstrated resistance to changing their favorite team. There was no overall difference between the three groups (see Table 4). There was a difference by gender within the preoperational group,  $\chi^2(1, N = 12) = 4.286, p < .05$ . The boys were resistant to change and the girls were not (because no girls had a favorite team). No significant difference was found based on household make-up, but there was a difference by ethnicity within the preoperational group,  $\chi^2(1, N = 12) = 4.286,$

$p < .05$ . None of the African American children were resistant to change because they did not have a favorite team; 57% of the Caucasian children were resistant to changing their favorite team.

Considering that one reason children gave for liking a specific team was because the team performed well (i.e., they won), one might expect low resistance to change. Eighty-five percent of the children who did have a favorite team, however, said that losing seasons or changes to the team would not alter their feelings about a favorite team. Cognitive complexity may provide one explanation for the resistance to change, particularly among the concrete operational children who gave more than one reason for liking a team. For a child who has a more complex cognitive structure (multiple reasons for liking a favorite team), a challenge to one dimension (performance) would likely not be sufficient to sever the attachment to a favorite team. Other dimensions would provide a basis for continuing to like the team.

Another explanation for the resistance to change is that a child's attachment to a team may not have been challenged. Some of the children, particularly the preoperational group, may not have been able to conceptualize a team's losing all of their games or changes to the team operations (a team moving or players changing to another team) because preoperational children are likely to focus on immediate perceptions rather than abstract ideas. Additional measures should be devised to measure a preoperational child's resistance to change.

### ***Psychological Commitment—Volition***

Fifty percent of the children who had a favorite team demonstrated volition. Examples of volitional responses included "I would like Notre Dame even if my dad did not," "I would like the team even if my dad and brother did not," "Even though my dad likes the Lakers, I like the Bulls better," and "I like the team even though none of my friends or family like them." Responses suggesting a lack of volition included "I like the team because my dad likes them," "My father and brother like Michigan so I like Michigan," "I would not like the Raiders if my father did not like them," and "I like the Reds because my cousin likes them."

There was no significant difference between the groups on the basis of volition (see Table 4). The only within-group differences were by gender,  $\chi^2 (1, N = 20) = 5.495$ ,  $p < .05$ , and race,  $\chi^2 (1, N = 20) = 6.282$ ,  $p < .05$ , within the transitional group. Sixty percent of the boys had a favorite team because of a personal choice, and 90% of the girls liked a specific team as a result of the direct influence of others. Seventy-one percent of African American children liked a team on the basis of a personal choice, whereas 85% of the Caucasian children were influenced by others to like a specific team. It is unclear why there was such a difference relative to volition and race.

### ***Summary of Psychological Commitment***

Ten of the children interviewed demonstrated psychological commitment to a specific college or professional sports team on the basis of the standards used in the assessment (see Table 4). In the concrete operational group, the two girls were 8 years old and the boy was 9 years old. In the transitional group, four of the boys were 8 years old and one was 5 years old, and the girl was 8 years old. The child characterized as preoperational was a 5-year-old boy. These results suggest that part of the answer to the first research question is that children at both phases of cognitive development and children who are transitioning are capable of demonstrating a psychological commitment to a sports team. The next portion of the interview assessed whether children at different stages of cognitive development demonstrated the behavioral consistency thought to characterize team loyalty.

### ***Behavioral Consistency—Attending Games***

A small number of children in each group did report that they had attended a game played by their favorite team (see Table 4). There was no significant difference between the three groups on the basis of attending games, nor were there any significant within-group differences (gender, race, and household make-up).

It is important to acknowledge several constraints that limit attendance as an indicator of a child's loyalty to a sports team. Parents (and other family members) influence whether children attend sporting events. Parental influence would include the actual decision to attend, providing the resources to purchase tickets, transportation to a game, and supervision at a game. Proximity is also a concern; if a child (or an adult) likes a team that is not local, the opportunity to attend games is obviously limited. Seventy-seven percent of the children interviewed had a favorite college or professional team that was nonlocal. Interestingly, half of the children who attended a game did watch their favorite team play in another city. These results suggest that attendance may be one indicator of behavior, but when individuals develop a commitment to a nonlocal team, or when there are constraints on attending, other behaviors should be measured to illustrate behavioral consistency.

### ***Behavioral Consistency—Watching Team on Television***

There was no significant difference between the preoperational, transitional, and concrete operational children in watching a favorite team on television. Thirty-four percent of all the children interviewed watched their favorite team on television (see Table 4). Closer examination of the responses indicated, however, that the preoperational children did not make a deliberate choice to watch their favorite team. Preoperational children said they watched their favorite team "if that is what my dad is watching" or "when my dad watches them I do too." The children in the transitional and concrete operational groups explained that they had made a specific choice to watch their favorite team on television.

There was a significant gender difference within the preoperational  $\chi^2(1, N = 12) = 4.286, p < .05$  and transitional groups,  $\chi^2(1, N = 20) = 9.899, p < .05$ , and also a difference in the preoperational group on the basis of ethnicity,  $\chi^2(1, N = 12) = 4.286, p < .05$ . No girls characterized as preoperational had a favorite team, and responses suggested that they did not watch any team on television. Eight out of 10 boys in the transitional group reported that they did watch their favorite team, but only 1 girl said that she watched her favorite team. The difference by race within the preoperational group was that no African American children watched their favorite team on television (because none reported having a favorite team).

### ***Behavioral Consistency—Reading About Team***

Reading about a favorite team has focused on reading a newspaper or magazine for information about a team (e.g., statistics, game summaries, stories about the team), (McPherson, 1976; Smith et al., 1981). Results from the pilot study showed that children aged 5–6 were still learning how to read. None of the 5–6-year-old children, however, were able to read a newspaper or the type of magazine generally associated with sports teams. The measure was maintained, however, in order to determine if children in the older group read about a favorite team.

Only 1 child characterized as transitional said that he did read a newspaper or magazine for information about his favorite team. The older children in the concrete operational group explained that they did read about sports in general or other sports topics; examples cited included fiction books, books with sports jokes, and books about sports records, and 1 child talked about a pricing guide for baseball cards. The latter two responses suggest that children in the concrete operational phase are capable of reading newspapers and sports magazines

and understanding the statistics in sports, but they may not choose to do so. Overall, reading about a favorite team was not important to the children interviewed.

### ***Behavioral Consistency – Talking About Favorite Team***

No significant difference was found between the groups on the basis of talking about a favorite team (see Table 4). Only a small percentage of children in each group said that they did talk about a favorite team with other people. No significant differences were found for race or household income, but there was a gender difference in the transitional group,  $\chi^2(1, N = 20) = 5.495, p < .05$ . Half of the boys said that they did talk about their favorite team with others, but only 1 girl (out of 10) reported similar behavior.

Talking about a favorite team, for the children interviewed, included discussing a team's overall performance (win/loss record) and the results of a recent game. When asked how often they talked about their favorite team, the typical response from the children was "sometimes." The concept of frequency was difficult for many of the children to grasp without specific examples (e.g., once a day, once a week, etc.). It was not clear if talking about a favorite team was simply part of an occasional conversation or if a favorite team was a child's primary topic of conversation. Some of the concrete operational children readily discussed their favorite team, suggesting that talking about a team may be a good indicator of behavioral consistency for children, depending on appropriate measurement.

### ***Behavioral Consistency—Team-Related Apparel***

Cialdini et al. (1976) suggested that wearing team apparel broadcasts an individual's identification with a team. Smith et al. (1981) held the same view that loyal fans do own and wear apparel representing their favorite team. People who are loyal to a specific team are not the only ones who wear team apparel; others may buy team-related clothing because of the style, attractive colors, or the current clothing fad. Among children, though, owning team-related items or, more important being aware of certain items, like a team shirt or hat, may serve as an indicator that loyalty is developing.

No significant difference was found between the stages of cognitive development and owning team-related apparel (see Table 4). There were no differences on the basis of race or household make-up, but there was a gender difference. In the transitional group,  $\chi^2(1, N = 20) = 5.000, p < .05$ , and the concrete operational group,  $\chi^2(1, N = 18) = 4.500, p < .05$ , boys did own apparel (e.g., T-shirt, sweatshirt, jacket, cap) representing their favorite team (see Table 4), but no girls owned apparel for their favorite team. It is not clear why no girls owned team apparel; some possible explanations include gender stereotyping (parents or the children thinking of team apparel as masculine) or girls' having a greater interest in more "fashionable" clothes.

Responses to the questions about specific apparel provided some support for the idea that wearing team apparel can be a good indicator of team loyalty in children. Children in all three groups who said that they did own team apparel talked about wearing an item (e.g., T-shirt or cap) to "show they are my favorite team" or because "they [apparel] show my favorite team."

### ***Summary of Behavioral Consistency***

Twenty-two of the children interviewed did demonstrate one or more of the measured behaviors (see Table 4), but the frequency or intensity of the behaviors did not reflect loyalty as commonly associated with the measures (McPherson, 1976; Smith et al., 1981). The children who attended games did so infrequently (once or twice), children did not fill significant portions of their days reading and talking about a favorite team (only 1 child read about his favorite team), and only a small number of children owned team-related merchandise.

Another point to consider besides the question of intensity is how many behaviors constitute loyalty to a team. One position is that all five behaviors should be demonstrated. Using this standard, none of the children interviewed would be characterized by behavioral consistency (see Table 4). The data from the interviews does provide support for the idea that children characterized by preoperational, transitional, and concrete operational thought are capable of demonstrating the behaviors that reflect team loyalty (Research Question 1). It is not likely, however, that children, particularly preoperational children, will demonstrate the intensity or frequency that would constitute behavioral consistency.

***Team Loyalty***

The results from the interviews suggest that children characterized by preoperational, transitional, and concrete operational thought are capable of demonstrating the psychological commitment and resistance to change indicative of team loyalty, but not necessarily the behavioral consistency. Responses from 1 of the preoperational, 6 of the transitional, and 3 of the concrete operational children suggested that they did have a psychological commitment to a sports team. Among those 10 children, responses from 5 boys (1 preoperational, 3 transitional, and 1 concrete operational) indicated that they also engaged in four of the five behaviors measured. Without a clearer distinction of the behavioral intensity and frequency appropriate for children, however, the results do not support the contention that children characterized by preoperational, transitional, and concrete operational thought are capable of demonstrating team loyalty.

***Influence of Socializing Agents***

The final portion of the interview protocol examined which socializing agents influenced a child to choose a particular team as her or his favorite. Each child was asked whether her or his family members and friends had a favorite sports team, if they attended or watched games with others (parents, siblings, or friends), and whether they talked with others about their favorite team.

***Knowing Family and Friends' Favorite Team***

Children in all three groups said that they knew whether family members or friends had a favorite team (see Table 5). In all three groups, more children reported knowing their father's favorite team than knowing their mother's, siblings', or friend's favorite team. The children's answers were further examined to find out if they had the same favorite team as a family member and/or friends. Among the children who did have a favorite team, 50% had the same favorite team as their father, 31% had the same favorite team as a sibling (primarily an older brother), 12% had the same favorite team as their mother, and 8% had the same favorite team as a friend. There was no significant difference between the groups relative to having the same favorite team as another family member or friend.

The results suggest that across the phases of cognitive development, fathers had an important influence on a child's selection of a favorite team. Older brothers also seemed to have an influence. Within the preoperational group, none of the girls knew whether family members or their friends had a favorite sports team. The girls at the preoperational level, primarily 5–6-year-old girls, seemed to have very little interest in sports spectating.

***Attending and Talking About a Favorite Team's Games***

The influence of fathers emerged very clearly relative to attending games. Only 1 child talked about going to games with a brother, and the remaining children who had attended a game all talked about going with their fathers. It could be argued that a father might take

TABLE 5 Know Family and Friends' Favorite Team

|                    | Preoperational |   |       |    | Transitional |   |       |    | Concrete operational |   |       |    | $\chi^2$ | <i>p</i>                     |
|--------------------|----------------|---|-------|----|--------------|---|-------|----|----------------------|---|-------|----|----------|------------------------------|
|                    | M              | F | Total | %  | M            | F | Total | %  | M                    | F | Total | %  |          |                              |
| Father's favorite  |                |   |       |    |              |   |       |    |                      |   |       |    |          |                              |
| Yes                | 4              | 0 | 4     | 33 | 7            | 4 | 11    | 55 | 4                    | 3 | 7     | 39 |          | 2.836 <sup>a</sup> <i>ns</i> |
| No                 | 3              | 5 | 8     | 67 | 3            | 6 | 9     | 45 | 4                    | 7 | 11    | 61 |          |                              |
| Mother's favorite  |                |   |       |    |              |   |       |    |                      |   |       |    |          |                              |
| Yes                | 1              | 0 | 1     | 8  | 2            | 3 | 5     | 25 | 1                    | 3 | 4     | 22 |          | 2.712 <sup>a</sup> <i>ns</i> |
| No                 | 6              | 5 | 11    | 92 | 8            | 7 | 15    | 75 | 7                    | 7 | 14    | 78 |          |                              |
| Sibling's favorite |                |   |       |    |              |   |       |    |                      |   |       |    |          |                              |
| Yes                | 3              | 0 | 3     | 25 | 3            | 2 | 5     | 25 | 4                    | 1 | 5     | 23 |          | 1.790 <sup>a</sup> <i>ns</i> |
| No                 | 4              | 5 | 9     | 75 | 7            | 8 | 15    | 75 | 4                    | 9 | 13    | 72 |          |                              |
| Friend's favorite  |                |   |       |    |              |   |       |    |                      |   |       |    |          |                              |
| Yes                | 1              | 0 | 1     | 8  | 4            | 1 | 5     | 25 | 2                    | 1 | 3     | 17 |          | 3.641 <sup>a</sup> <i>ns</i> |
| No                 | 6              | 5 | 11    | 92 | 6            | 9 | 15    | 75 | 6                    | 9 | 15    | 83 |          |                              |

Note. M = male; F = female.

<sup>a</sup>No overall difference was found between the groups.

*df* = 2; *N* = 50.

a child to a game because he thinks the child would enjoy the outing and to spend time with the child. Considering the cost of attending college and professional sporting events, it is more likely that a father attends a game with a child because both share a common interest in the team, or a father has an interest in a team that he wants to share with his child. A similar influence from fathers was found among those children who talked about their favorite teams. Two children said they talked with their brother, 3 talked with a friend, and 8 talked with their father about their favorite teams.

### ***Watching a Favorite Team's Games***

The responses to the questions about watching a favorite team also indicated a strong influence from fathers. Seventy-six percent of the children who watched their favorite team on television said that they watched either "with my dad" or "when my dad watches a game." One child said she or he watched the game when "my brother watches," and 2 children said they watched their favorite team but were not clear on whether others watched with them. The responses from the children suggested that fathers influenced a child by talking about a team, attending a game, or watching a team on television. Siblings, particularly older brothers, seemed to have some influence, but it is unclear from the interviews whether an older brother introduced a child to a favorite team or reinforced the importance of a particular team already introduced by a child's father.

The interview responses indicated that family members, particularly a child's father, exerted a greater influence on a child's attachment to a favorite team than did friends (Research Questions 2 and 3). For all groups, a higher percentage of children knew their father's favorite team and liked the same team as their father compared with a friend. The responses also indicated that if children attended a game, watched a favorite team on television, or talked about their favorite team, they did so with their fathers rather than their friends. A question for future research is whether friends exert a greater influence at later points in an individual's life (e.g., adolescence).

### ***Parental Survey***

It is interesting to note that mothers were not associated with attending a game or watching a favorite team on television. Among the children interviewed, there was an overwhelming dissociation between mothers and sports teams. When asked if their mother had a favorite sports team, children in all three groups said that their mothers did not have a favorite team, and many went on to explain that "she doesn't like sports" or "she is not interested in sports." The findings from the interviews were particularly interesting when compared with the responses from the parental survey.

All the mothers and fathers of the children who had a favorite team reported that they did have a favorite sports team. Both parents also reported that they did watch their favorite team with their son or daughter. Responses from the interviews, however, indicated that none of the children associated watching any sports with their mother. Seventy-three percent of the fathers and 65% of the mothers said that they did talk to their son or daughter about their favorite team, but none of the children associated talking about a favorite team with their mother. Even though mothers reported having a favorite sports team, that they watched the team on television, and that they talked about the team with their son or daughter, the children associated sports teams with fathers and brothers. Across gender, age, and phase of cognitive development the association of sports with fathers was the same. A question that bears further consideration is whether children at the preoperational and concrete operational phases of development (or perhaps children under age 10) do not associate sports with mothers specifically, or if they do not associate sports with women in general.



## Discussion

Sport spectating is a predominant form of leisure behavior, as evidenced by the millions of people who attend and watch games (Badenhausen et al., 1997; Madrigal, 1995) and the loyalty that people have to sports teams (Kolbe & James, 2000). Previous research on loyalty helps us to characterize the behavioral aspect of loyalty and to assess whether people have a psychological commitment to sports teams (Backman & Crompton, 1991; Day, 1969; Howard et al., 1988; Jacoby & Chestnut, 1978; Pritchard et al., 1992). Research has not, however, addressed the question of how people initially become interested in and develop loyalty to specific teams.

The present study was conducted by using the conceptual framework proposed by Iwasaki and Havitz (1998) to address the question of when and how loyalty develops. Specifically, the study investigated the moderating influence of cognitive development (a personal moderator) and four socializing agents (father, mother, sibling, and television) on the development of psychological commitment and resistance to change to ascertain when children are first capable of demonstrating team loyalty.

Children characterized by preoperational, transitional, or concrete operational thought did have preferences for a specific sports team and were able to demonstrate psychological commitment, resistance to change, and some behaviors indicative of loyalty, but not necessarily the consistency generally associated with loyalty (McPherson, 1976; Smith et al., 1981). The results suggest that children are capable of developing commitment to a sports team early in life (as young as age 5 in the current study), but likely will not demonstrate the behavioral consistency associated with team loyalty till later in life (beyond age 9 for the children interviewed). The question of whether a child, particularly a young child (7 and under) may be considered loyal will depend on whether children have opportunities to engage in particular spectating behaviors (e.g., attending games, watching games on television, talking about favorite team, etc.). Implications from the current study pertaining to the different elements of the conceptual model are discussed below.

### *Cognitive Development*

A criticism of Piaget's (1970) theory of cognitive development has been its inability to appropriately identify children who do not "fit" within a specific phase of development. Cognitive development is recognized as a dynamic process (Thelen & Smith, 1994), but instead of trying to better understand the movement between stages, the theory is often discounted. One contribution from the current study is including the concept of transitioning to better understand a child's progression from preoperational to concrete operational thinking.

The counting strategy used by children in the current study is an example of transitioning between phases of development. Children acknowledged that the rows of pennies did not look the same; their counting the pennies illustrates that instead of comparing the conditions (abstract thinking and reasoning), the children focused on their immediate perceptions. The children were moving beyond preoperational thinking, but they had not fully attained concrete operational thinking—they were transitioning.

Future research examining levels of cognitive development should include the concept of transitioning to better understand children's cognitive development. The current study did not find examples of transitioning relative to conservation of mass and volume, but it is reasonable to expect that similar patterns of reasoning could be identified. Being able to characterize and distinguish between the reasoning and thinking of children at different phases of development will allow researchers to examine the influence of different socializing agents on children's development at different phases and during the transition between phases.

*Involvement*

Half of the children interviewed said that they did have a favorite sports team, suggesting that they had an interest in or attraction to a specific team. Only children who had a favorite sports team demonstrated a psychological commitment that was resistant to change. The results support the contention of Iwasaki and Havitz (1998) that involvement is an important antecedent of psychological commitment. The current study used a very simple measure of attraction or interest; future research should consider additional measures of involvement in the study of team loyalty. Two dimensions that should be looked at more closely are centrality and sign value. The more central (or important) a team is to an individual, the stronger the psychological commitment should be.

*Psychological Commitment*

*Cognitive Complexity*

The current study is the first effort to assess a child's psychological commitment to a sports team. The findings, though instructive, raise several questions that may be considered through future research. Scholars have proposed that psychological commitment is characterized by informational complexity and consistency, volitional choice, position involvement, and resistance to change (Buchanan, 1985; Crosby & Taylor, 1983; Pritchard, 1992). The current study extends previous research by measuring three dimensions of commitment—cognitive complexity, volitional choice, and resistance to change—with varying degrees of success.

Concrete operational and transitional children demonstrated cognitive complexity by offering two or more reasons for having a favorite team. Preoperational children consistently gave only one reason for liking a favorite team, which may or may not reflect complexity. A question that researchers should address is “what constitutes complex thinking for a child?” It can be argued that if a child likes a team because they win, there is recognition of the positive value associated with winning, which could be a very complex thought for a 5-year-old. It is my position that the ability to express a reason for having a favorite team that goes beyond “because I like them” does demonstrate cognitive complexity for a young child (age 7 and under). In addition to wrestling with what constitutes cognitive complexity for children, additional research should also work to assess informational consistency and position involvement, not directly examined in the current study.

*Resistance to Change*

Another question raised by the current study is the extent to which children are resistant to changing their favorite team. Concrete operational and transitional children did seem to understand the implications of the questions used in the study. It was not clear if the preoperational children were able to imagine the circumstances described (e.g., a team losing all its games), considering that a child at that phase focuses primarily on immediate perceptions. Future research should consider additional ways to assess resistance to change, potentially through longitudinal studies that can track preference stability.

*Volition*

Children in the transitional and concrete operational groups did understand the concept of volitional choice. Responses such as “Even though my dad likes the Lakers, I like the Bulls better” and “None of my friends like the same team I do, but that doesn't matter” indicate that a child had freely chosen a specific team as his or her favorite. It is questionable, though, whether the preoperational children understood the concept. One child in the preoperational

group said that he would like a favorite team even if other people did not like it. The child did like the same team as his father and brother, and it is unclear if he could truly imagine liking a different team than his father. Future research should consider whether there are more indirect approaches for assessing volition for younger children and whether a preoperational child can assume responsibility for his or her choices.

### ***Behavioral Consistency***

One question to consider is whether the behaviors commonly associated with team loyalty (McPherson, 1976; Smith et al., 1981) are applicable to children. Several constraints limit the utility of the behaviors when measuring a child's loyalty to a sports team. Parents (and other family members) influence whether children attend sporting events and whether they watch sporting events. Having a favorite team located in another city limits the ability to attend and watch games. A child's ability to read will influence whether he or she reads newspapers and sports magazines. Children may not have the financial resources to purchase team apparel or may be restricted in their choice of apparel by parents. Researchers should consider whether other behaviors would be more realistic indicators of a child's loyalty to a sports team, or whether it is reasonable to suggest that children are not capable of demonstrating loyalty to a sports team because of the lack of behavioral consistency.

Iwasaki and Havitz (1998) proposed that loyal behaviors should reflect duration, frequency, intensity, sequence, proportion, and probability. The behaviors measured in the current study lacked these characteristics. A question to consider for sport spectating is what constitutes duration and frequency. How many games should a person attend or watch on television during a season? Smith et al. proposed that "significant portions" of one's day would be filled with talking and reading about a team. What constitutes a significant portion? In addition, how many behaviors must an individual demonstrate to reflect loyalty to a team? One position is that all five behaviors from the current study should be demonstrated. One approach to measuring behavioral consistency might be to identify key behaviors (e.g., what do sport spectators do), identify potential constraints to engaging in key behaviors, and develop a weighted scale accounting for different constraints. At the least, future research should begin to address whether and how the number and intensity of behaviors that constitute team loyalty are the same or different for adults and children.

### ***Socializing Agents***

#### ***Family***

Previous research on leisure preferences (Crawford et al., 1986; Iso-Ahola & Mannell, 1985; Kleiber & Durkin, 1985) and leisure and sports socialization (Hoff & Ellis, 1992; Iso-Ahola, 1980; Kenyon & McPherson, 1973; McPherson, 1976) have shown that family members and friends are important socializing agents early in life. The current study extends our understanding of leisure and sports socialization relative to sports spectating. Fathers were found to have exerted a primary influence on children's developing a preference for and commitment to a sports team. Children were more likely to know their father's favorite team compared with knowing other family members' and friends' favorite teams, and children talked about and watched a favorite team on television with their fathers.

Another question to consider is whether socializing agents operating concurrently have individual, additive, or conflicting influences on an individual and whether one agent has a primary influence. No responses from the children interviewed suggested that television was a primary socializing agent, introducing children to specific teams. Responses did suggest, however, that television served an important role by reinforcing a child's attachment to a

specific team by providing opportunities to watch a team. Future research should investigate whether television is a primary socializing agent or whether it reinforces the lessons from another agent (e.g., watching TV with dad teaches a child about dad's favorite team).

The extent to which children view sports spectating as a male versus a female activity should also be investigated. All the fathers and mothers in the study reported that they had a favorite sports team and that they talked about and watched their favorite team with their child. Responses from the children, however, suggested that sport teams were associated with fathers and not mothers. Several children went so far as to say that their mothers were not interested in sports. Fathers in the current study had a stronger influence on a child's selection of a favorite sports team but it is not clear why. One reason may be the perpetuation of sports stereotyping, at least among younger children (under age 10). Even though research has demonstrated that mothers are primary decision makers regarding children's participation in activities (Howard & Madrigal, 1990), sports spectating may still be thought of as a male activity.

### ***Limitations and Future Research***

As with all research, there are several limitations that must be acknowledged. Time constraints as well as manpower constraints (only one interviewer) dictated that sample size ( $N = 50$ ) be kept manageable. The 50 interviews did not provide a representative sample in terms of generalization but did provide the information needed to examine the four research questions.

The interview protocol was used for the first time as an initial effort to examine the development of loyalty. As with any dynamic process, there are modifications that can be made in future research. For example, additional questions should be included to examine in more detail the influence of television; more indirect questions could be used to assess volition and resistance to change. Questions are also needed to examine more fully the differential role of friends at different phases of cognitive development. Loyalty to other leisure activities or programs should also be examined.

The current study highlights the importance of conducting longitudinal research to better understand the development and persistence of team loyalty. Research with children should begin very early (at least by age 3) to ascertain the influence of family members on a child's initial socialization to sports spectating. Of particular interest is how (or whether) parents and other family members introduce a child to a sports team and how a parent communicates to a child the value of following a sports team. On a less positive note, it is possible that children learn early in life to devote their time and resources to following a sports team, which could be detrimental to future family relations (e.g., putting the team before family activities).

As with any leisure activity, the devotion of time and resources to a sports team can be carried to an extreme. Loyalty can reach a point where an individual becomes so obsessed or fanatical that devotion to a team can result in violence, as evidenced by episodes of hooliganism and riots. Longitudinal studies of the development of loyalty may provide insights into why a strong commitment translates into deviant behavior. Research may illustrate whether an individual has other relationships in his or her life that provide a sense of balance or if one's esteem is derived so completely from a single relationship (commitment to a sports team) that something as simple as losing a game could incite a person to violence.

Longitudinal research is also important in order to more accurately assess the stability of an individual's commitment to a specific team and the persistence of loyalty. One question that has not been considered is how long an individual has to follow a team in order to be

considered loyal. Another question to consider is whether loyalty may wane over time, and, if so, what would weaken an individual's commitment to a specific team. The current study does provide support for portions of the conceptual framework proposed by Iwasaki and Havitz (1998), particularly the relationship between involvement (attraction/preference), psychological commitment, and resistance to change. Future research should continue to explore the utility of the framework in an effort to better understand when and how loyalty develops.

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